

Sagar Sinha

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EDUCATION

MS Computer Science

Arizona State University

Dec 2025

GPA: 3.93/4.0

- **Relevant Coursework:** Data Mining, Applied Regression Analysis, NLP, Data Visualization, Distributed Database Systems

BS Computer Engineering

Pandit Deendayal Energy University

May 2022

GPA: 9.02/10.0

- **Relevant Coursework:** Data Structures, Algorithms, RDBMS, Data Warehousing, Statistics, Machine Learning

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, Bash
- **Libraries:** Pandas, NumPy, Scikit-learn, PyTorch, Statsmodels, spaCy, Gensim, fastText, NLTK, Matplotlib, Seaborn, Plotly, ggplot2, Selenium, Airflow, PySpark
- **Platforms:** Tableau, Looker Studio, MS Excel, dbt, Docker, MySQL, PostgreSQL, Google Cloud, Snowflake, MongoDB

PROFESSIONAL EXPERIENCE

Data Engineer Intern, Mindsparkd

Mar 2026 – Present

- Engineered a **System Health data pipeline** using **AWS Lambda, DynamoDB, Aurora PostgreSQL, API Gateway**, and **CloudWatch** to transform Kairos interaction logs into rolling-window anomaly insights.
- Improved high-frequency data retrieval performance by **40%** by optimizing **Aurora PostgreSQL** query paths, removing redundant joins, and adding targeted indexes for faster analytical access.
- Audited **serverless data workflows** across Lambda, Aurora, and DynamoDB using **CloudWatch Insights, RDS metrics**, and **Performance Insights** to identify retry, timeout, read/write, and scaling risks.
- **Technical Skills:** Python, SQL, AWS Lambda, Aurora PostgreSQL, DynamoDB, API Gateway, Step Functions, CloudWatch, Performance Insights, Terraform, Git.

AI & Data Processing Externship, Extern

Sep 2025 – Present

- Built a modular, AI-powered pipeline to process 200+ page mortgage blob files—combining OCR (Tesseract, PaddleOCR), PDF parsing (PyMuPDF), and RAG techniques for intelligent data extraction, classification, and search.
- Engineered a RAG-based retrieval architecture via LlamaIndex; optimized multi-document ingestion using semantic chunking and metadata-aware indexing while benchmarking Mistral and Phi-2 performance.
- Benchmarked E2E pipeline performance—targeting OCR accuracy, RAG retrieval metrics, and routing logic—to deliver technical trade-off analyses, deployment roadmaps, and a functional UI demo.
- **Technical Skills:** RAG(LlamaIndex), LLMs(Mistral, Phi-2), OCR(Tesseract, PaddleOCR), Vector Databases, Semantic Chunking, Metadata Filtering, PDF Parsing(PyMuPDF)

Data Specialist, Part Time, Arizona State University

Jul 2024 – Present

- Engineered an Airflow-orchestrated ELT pipeline on AWS to ingest 50K+ Canvas records; automated extraction via Selenium and BeautifulSoup to S3, with structured loading into Redshift for centralized modeling.
- Benchmarked health surveys (ACHA, WISHES) against national datasets, identifying key trends to enhance wellness initiatives.
- Streamlined data collection and analysis procedures with analysts and health coordinators, improving report consistency and accelerating decisions on student wellness programs.
- **Technical Skills:** Python, MySQL, Tableau, Amazon Web Services (AWS), Exploratory Data Analysis, Data Modeling

Data Researcher, Accredian

Apr 2023 – Jul 2023

- Revamped ML/NLP modules (classification, GRU, LSTM, BERT), improving clarity and applicability for 3,000+ learners.
- Resolved 90% of technical tickets on time by fixing ML errors, clarifying concepts, and troubleshooting environments, reducing escalations to senior instructors.
- **Technical Skills:** Python (Pandas, NumPy, Torch), Feature Engineering Pipelines, Feature Engineering, Production Support, System Optimization, Root Cause Analysis

RELEVANT PROJECTS

Enterprise Sales Data Warehouse

- Architected a SQL Server data warehouse using a **Medallion framework (Bronze, Silver, Gold)** to consolidate disparate ERP and CRM CSV sources into a centralised **Star Schema**, creating a single point of truth for cross-functional reporting.
- Automated end-to-end **ETL pipelines** via **SQL Stored Procedures and bulk inserts**, transforming manual data collection into automated refreshes that reduce reporting latency from weeks of manual preparation to minutes of processing time.
- Optimised **data integrity and governance** by implementing advanced cleansing logic and surrogate keys, achieving **100% record accuracy** through automated quality checks and establishing full **data lineage** for all business-ready views.
- **Technical Skills:** Database Architecture Design, Metric Development, ETL architecture, Snowflake SQL

Distributed Banking System

- Containerized and configured a distributed, **sharded MongoDB** cluster using **Docker**, sustaining up to **200K+ concurrent requests** with **sub-70 ms read latency** under load.
- Simulated heavy read/write workloads and implemented automated **replica-set health checks** and basic failover monitoring, maintaining **>99% successful requests** at peak concurrency.

- **Technical Skills:** Databases, Data Modeling, Database Consistency, Concurrency Management, Golang, Modular Architecture, Containerization, Routing

Urban Mobility Insights

- Engineered an end-to-end GCP pipeline using **Terraform, Airflow, and BigQuery** to ingest and partition NYC taxi data into a scalable warehouse.
- Built a **dbt-based dimensional model** and **Looker Studio dashboard** to visualize trip trends and revenue, applying rigorous data testing and documentation.
- **Technical Skills:** Dimensional Data Modeling, Google Cloud (Google Cloud Storage, Google Bigquery, Dataproc Serverless)

ACHIEVEMENTS

- **ASU AI Spark Challenge:** Won 2nd Prize and a \$2,000 scholarship at the hackathon for co-developing Socratic AI, a platform teaching GPU-accelerated Python coding through interactive dialogue and live benchmarking.
- **SunHacks 2024, ASU:** Won the MLH Best Use of Midnight Award for developing Hack Thy Neighbour, a decentralized lending platform using blockchain and smart contracts to enable transparent, community-governed fund pooling.